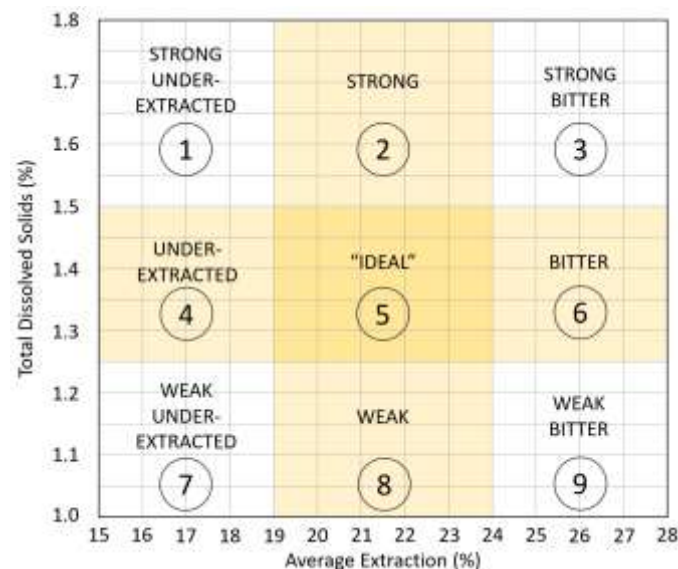


金杯理論

如何定義一杯好喝的咖啡？1960 年咖啡沖煮學會(CBI)推出咖啡沖煮控制圖，如下圖所示。認為咖啡的味道應與沖煮強度(Strength)、咖啡萃取率(Average Extraction)有著密切的關係，並且訂出一些準則能讓沖煮咖啡在調整風味上有所依據，達到出杯品質的穩定和一致性。根據此咖啡沖煮控制圖可以將一杯咖啡分成下列 9 種品質分級：

- (1) 強且萃取不足：萃取率<19%，1.5%<沖煮強度
- (2) 強：19%≤萃取率≤24%，1.5%<沖煮強度
- (3) 強且苦：24%<萃取率，1.5%<沖煮強度
- (4) 萃取不足：萃取率<19%，1.25%≤沖煮強度≤1.5%
- (5) 理想且均衡：19%≤萃取率≤24%，1.25%≤沖煮強度≤1.5%
- (6) 苦：24%<萃取率，1.25%≤沖煮強度≤1.5%
- (7) 弱且萃取不足：萃取率<19%，沖煮強度<1.25%
- (8) 弱：19%≤萃取率≤24%，沖煮強度<1.25%
- (9) 弱且苦：24%<萃取率，沖煮強度<1.25%



濾過萃取的咖啡充煮方式是讓 86°C ~93°C 的沖煮水 BW (以 g 為單位) 不斷注入至濾紙上的乾燥咖啡粉 D (以 g 為單位)，此時咖啡粉會釋放咖啡化合物 M_{bev} (以 g 為單位) 和通過濾紙的充煮水混合而成咖啡液 BEV (以 g 為單位)。我們可以利用咖啡濃度計去量測咖啡的沖煮強度 TDS (以%為單位)，表示為溶解出的咖啡化合物重量 M_{bev} 在咖啡液 BEV 的比例

$$TDS(\%) = \frac{M_{bev}}{BEV}$$

咖啡液重量 BEV 應該將沖煮水加總咖啡化合物之外，還需考慮扣除在濾紙上咖啡粉吸水的重量。液體保留率 LRR 被定義每公克的咖啡粉吸收多少沖煮水重量的比率。因此咖啡液重量 BEV 可以寫成

$$BEV = BW + M_{bev} - LRR \times D$$

咖啡萃取率可以定義成乾燥咖啡粉與溶解出的咖啡化合物的比例，如下形式：

$$EXT(\%) = \frac{M_{bev}}{D}$$

請撰寫一個程式，根據輸入的乾燥咖啡粉重量、沖煮水重量、 LRR 和 TDS 值，計算其萃取率和咖啡液重，並且判斷咖啡的品質級數。

輸入說明：

第 1 行為 1 個整數，代表有 k 組測試資料，其中 $1 \leq k \leq 100$ 。

第 2 行至第 $2+k$ 行，每行資料依序為乾燥咖啡粉重量、沖煮水重量、 LRR 和 TDS 。

輸出說明：

根據每行輸入資料，分別計算咖啡萃取率、咖啡液重和咖啡品質的編號(1-9)，其中萃取率和咖啡液重四捨五入取至小數點以下第 2 位。

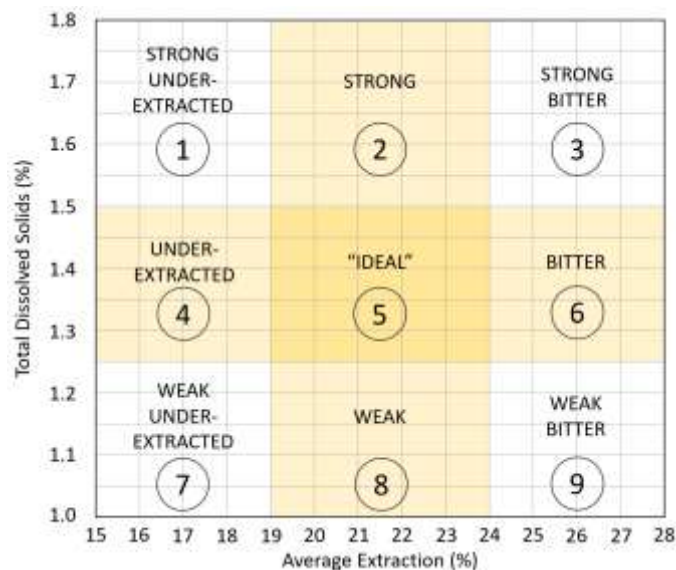
範例說明：

範例 1 輸入	範例 1 輸出
2 20.00 300.00 2.62 1.35 15.00 200.00 2.18 1.55	16.94 250.99 4 17.56 169.93 1
範例 2 輸入	範例 2 輸出
3 11.33 180.00 2.01 1.44 9.89 180.00 1.50 1.44 15.33 300.00 2.44 1.66	20.27 159.52 5 24.40 167.58 6 28.91 267.03 3

Gold Cup Theory

How do you define a good cup of coffee? In 1960, the Coffee Brewing Institute (CBI) introduced the Coffee Brewing Control Chart. They believed that the taste of coffee is closely related to its brewing strength (Strength) and coffee extraction rate (Average Extraction). They established guidelines to provide a basis for adjusting the flavor of brewed coffee, aiming to achieve stability and consistency in cup quality. Based on this Coffee Brewing Control Chart, a cup of coffee can be classified into the following 9 quality grades:

- (1) Strong & Under-extracted: Extraction < 19%, Strength > 1.5%
- (2) Strong: 19% <= Extraction <= 24%, Strength > 1.5%
- (3) Strong & Bitter: Extraction > 24%, Strength > 1.5%
- (4) Under-extracted: Extraction < 19%, 1.25% <= Strength <= 1.5%
- (5) Ideal & Balanced: 19% <= Extraction <= 24%, 1.25% <= Strength <= 1.5%
- (6) Bitter: Extraction > 24%, 1.25% <= Strength <= 1.5%
- (7) Weak & Under-extracted: Extraction < 19%, Strength < 1.25%
- (8) Weak: 19% <= Extraction <= 24%, Strength < 1.25%
- (9) Weak & Bitter: Extraction > 24%, Strength < 1.25%



The drip extraction brewing method involves continuously pouring brewing water BW (in g) at 86 degrees C to 93 degrees C onto dry coffee grounds D (in g) placed on a filter paper. During this process, the coffee grounds release coffee compounds M_{bev} (in g), which mix with the brewing water passing through the filter paper to form the coffee beverage BEV (in g). We can use a coffee refractometer to measure the brewing strength TDS (in %), which represents the ratio of the dissolved coffee compounds' weight M_{bev} to the coffee beverage BEV:

$$TDS(\%) = \frac{M_{bev}}{BEV}$$

The coffee beverage weight BEV should not only be the sum of the brewing water and coffee compounds, but it also needs to account for the deduction of the water weight absorbed by the coffee grounds on the filter paper. The Liquid

Retained Ratio (LRR) is defined as the ratio of brewing water weight absorbed per gram of coffee grounds. Therefore, the coffee beverage weight BEV can be written as:

$$EBEV = BW + M_{bev} - LRR \times D$$

The coffee extraction rate can be defined as the ratio of dissolved coffee compounds to the dry coffee grounds, in the following form:

$$EXT(\%) = \frac{M_{bev}}{D}$$

Please write a program that calculates the extraction rate and coffee beverage weight, and determines the coffee quality grade, based on the input values of dry coffee grounds weight, brewing water weight, LRR, and TDS.

Input Specification: Line 1 contains an integer representing k sets of test data, where $1 \leq k \leq 100$. Lines 2 to 2+k: Each line sequentially contains the dry coffee grounds weight, brewing water weight, LRR, and TDS.

Output Specification: Based on each line of input data, calculate the coffee extraction rate, coffee beverage weight, and the coffee quality number (1-9) respectively. The extraction rate and coffee beverage weight should be rounded to two decimal places.

Example Demonstrations:

Example 1

Input:

```
2
20.00 300.00 2.62 1.35
15.00 200.00 2.18 1.55
```

Output:

```
16.94 250.99 4
17.56 169.93 1
```

Example 2

Input:

```
3
11.33 180.00 2.01 1.44
9.89 180.00 1.50 1.44
15.33 300.00 2.44 1.66
```

Output:

```
20.27 159.52 5
24.40 167.58 6
28.91 267.03 3
```